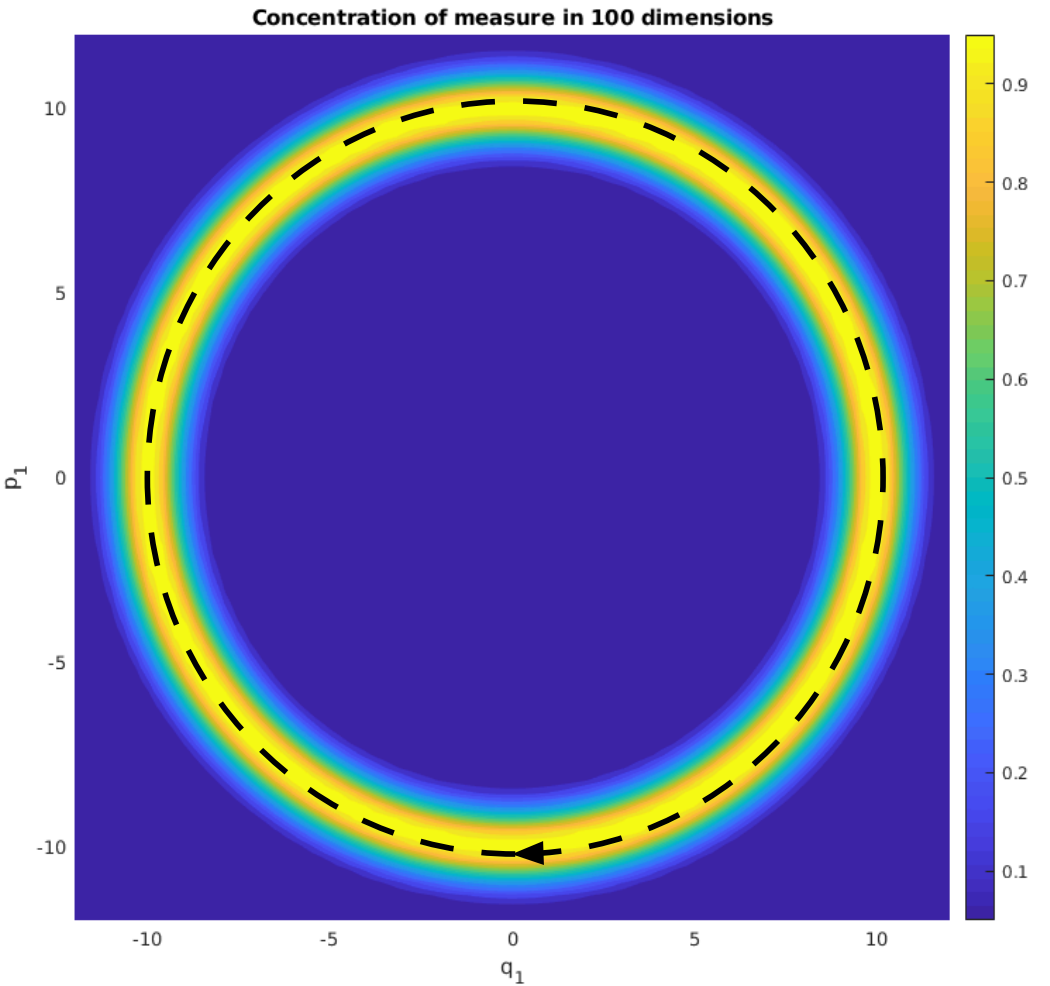
CONCENTRATION OF MEASURE AND HMC

22

- In general, if the Hessian of the target potential satisfies that $\lambda I \preceq \nabla^2 U(x) \preceq LI$ for some $0 < \lambda < L < \infty$, (strongly convex and smooth potential), and $H_{\min} := \inf_z H(z)$, then it is possible to show that

$$\mathbb{P}\left(\left|\sqrt{H(z)-H_{\min}}-\mathbb{E}\sqrt{H(z)-H_{\min}}\right|\geq t\right)\leq C\exp\left(-\frac{t^2}{C}\right)$$

- Hamiltonian is close to constant high probability density area



- HMC is very efficient in exploring this potentially complicated modes automatically.

- Does not apply to multi-modal distributions

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$$\mathbb{P}\left(\left|\sqrt{H(z) - H_{\min}} - \mathbb{E}\sqrt{H(z) - H_{\min}}\right| \geq t\right) \leq C \exp\left(-\frac{t^2}{C}\right)$$

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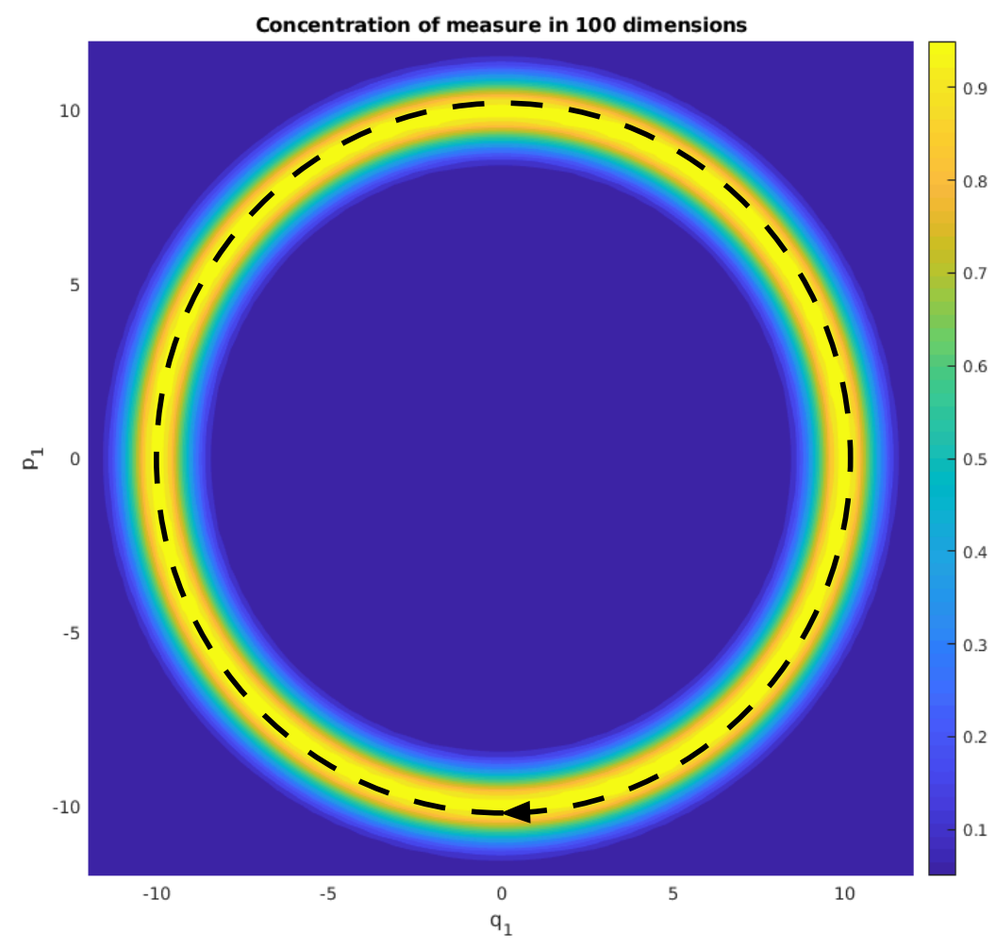

SUMMARY OF LOCAL SAMPLERS